

Internship report

Causal Sets Theory

Toward a construction of the Quantum Sequential Growth

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Abstract

Quantum gravity is one of the biggest riddle of our time. One approach to it is the Causal Set Theory (CST) that suggests a discrete structure of space-time at its fundamental level and aim to unify the framework of General Relativity and Quantum Mechanics by the concept of causality. This report is a review of the literature that cover details on the development of a dynamics in this field, with a focus on the sequential growth paradigm. Our main goal is here to study the construction of observables out of the dynamics. Starting by the classical case, we will then give some aspect of the work that currently occur in the understanding of the quantum case. When it is necessary, the reader could refer to a summary of useful vocabulary in the appendix A, as well as proofs of certain propositions in the appendix B. This content is essential, but has been included in appendix so as not to overburden the main reasoning.

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1 Introduction

The interaction of strong gravitational fields with quantum fields requires a theory of quantum gravity. One framework for quantum gravity is causal set theory, which replaces the classical spacetime continuum by the discrete structure of a network of elements at very small length scales. Ordinarily, one thinks of a space-time as a topological manifold called $\mathcal{M}$. It is endowed with a differentiable structure, with respect to which a metric g is defined. Unlike Riemannian spaces, spacetime has a Lorentzian signature $(-, +, +, +)$ which gives rise to local light cones and an associated global causal structure. This causal structure is defined as a **partially ordered set** : a pair $(\mathcal{M}, \prec)$ with $\prec$ denoting the causal ordering such that :

$$\forall x, y, z \in \mathcal{M}, \begin{cases} \text{Transitivity : } x \prec y \text{ and } y \prec z \implies x \prec z \\ \text{Acyclicity : } x \prec y \text{ and } y \prec x \implies x = y \\ \text{Reflexivity : } x \prec x \text{ (depend of the convention used).} \end{cases} \quad (1)$$

From the partially ordered set, we can construct a discretized version called **causal set** $(c, \prec)$ by adding the property of local finiteness. $(c, \prec)$ is a causal set if and only if it is a partially ordered set such that :

$$\forall x, y \in c, |\{z \in c | x \prec z \prec y\}| < \infty. \quad (2)$$

A finite region of a causal set contains a finite number of elements, which represent a notion of "space-time atoms". We discretize a finite region of space-time in two ways : by discretizing the causal order we obtain a causal set and by discretizing the hyper-volume of the region we obtain a number ε . Causal set theory (CST) as proposed in Bombelli et al. [3], takes the problem the other way and takes the causal structure as a crucial starting point to quantization, and claim that continuous space-time is just the approximation of a manifold-like causal set. The figure 1 resume this.

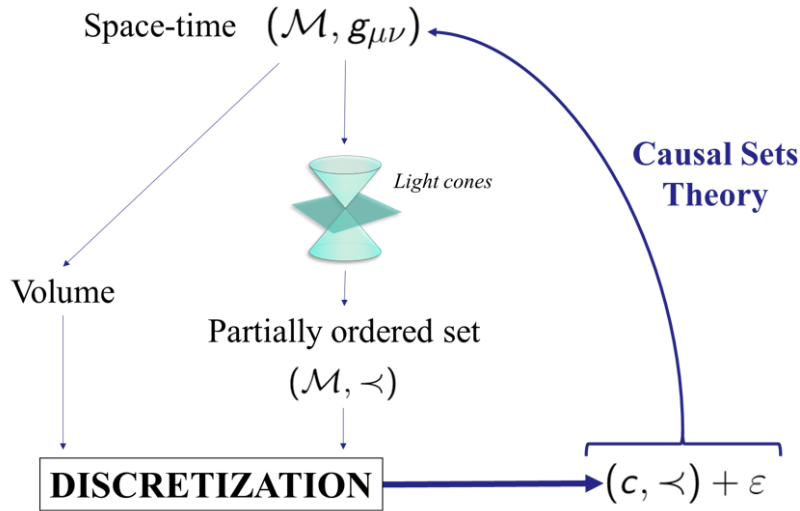


Figure 1: Causal Set Theory goal.

Why do we need ε ? Knowing (1), a fundamental theorem yields as follows.

Hawking–King–McCarthy–Malament–Levichev (HKMML) theorem :

Let $(\mathcal{M}_1, g_1)$ and $(\mathcal{M}_2, g_2)$ be d -dimensional Lorentzian manifolds with $d > 2$ such that the chronological (or timelike) past and future of each point in space-time is unique.

If $\exists f : (\mathcal{M}_1, \prec_1) \rightarrow (\mathcal{M}_2, \prec_2) | \forall x, y \in \mathcal{M}_1, x \prec_1 y \Leftrightarrow f(x) \prec_2 f(y)$ (so f is a causal bijection)

then $\exists F : (\mathcal{M}_1, g_1) \rightarrow (\mathcal{M}_2, g_2) | F_*(g_1) = \Omega^2 g_2$ (so F is a conformal bijection and F_* its push forward) and the two space-times are conformally isometric.

This means that the causal structure determine not just one space-time but the full conformal equivalence class of it. In other words, if we know only $(c_N, \prec)$, with c_N a causal set of N elements, we cannot recover a volume of space-time out of it. We lack the information contained in ρ (or equivalently ε) such that :

$$\frac{N}{\rho} = \int_V \sqrt{\det g} \, d^d x. \quad (3)$$

A causal set could be represented by a Hasse diagram like in figure 3. In this view, $b \prec a$ means that a is above b and that there is a causal line connecting b to a . In figure 3, we can see that $c \prec b \prec a$ and $d \not\prec a$. Here, $\{b, a\}$ is a link or a 2-chain, $\{c, b, a\}$ is a 3-chain of, $\{d, a\}$ is a 2-antichain and $\text{Past}(a)$ is the set of all element e such that $e \prec a$.

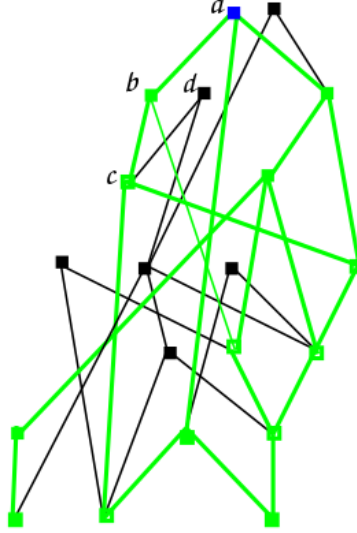


Figure 2: A causal set, the green set is $\text{Past}(a)$ [1].

The proto-causality encoded in $\prec$ give rise to causal order in the continuous approximation. The number of space-time atoms in a finite region is proportional to the space-time volume in the continuous limit. This idea is summed up by the CST slogan [4]:

$$\text{Order} + \text{Number} \sim \text{Lorentzian Geometry}. \quad (4)$$

We can see by a combinatorial analysis that most causal sets of cardinality n are what is called a Kleitman-Rothschild (or KR) causal set [5]. As depicted in figure 1, the KR structure contains only three layers. Half of the element are in a middle layer in the future of the bottom and in the past of the top layer, and the other half of the elements is equally divided between the top and bottom layers.

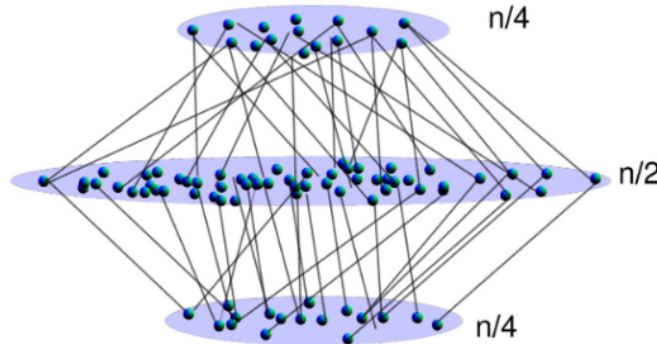


Figure 3: A KR causal set [4].

In these causal sets, the maximal length for a chain of time-like events is 1. It is obvious that we cannot have a smooth Lorentzian manifold with this structure. A causal set $(c, \prec)$ is manifold-like if :

- We can go from $(c, \prec) + \varepsilon$ to $(\mathcal{M}, g)$ by a **faithful embedding map**. Since the causal set approximating $(\mathcal{M}, g)$ is locally finite, it is a proper subset of $\mathcal{M}$. The injective map $\Phi : c \rightarrow (\mathcal{M}, g)$ is called an embedding map if and only if $x \prec_c y \Leftrightarrow \Phi(x) \prec_M \Phi(y)$. This map is faithful if the embedded points are distributed uniformly with unit density; and if the characteristic length over which the continuous geometry varies appreciably is everywhere much greater than the mean spacing between embedded points. In this way, a covariant space-time can be approximated from a manifold-like causal set.
- We can construct $(c, \prec) + \varepsilon$ from $(\mathcal{M}, g_{\mu\nu})$ by randomly select position in space-time with the Poisson distribution, so that the probability of finding n elements in a spacetime region of volume V is given by :

$$P_V(n) = \frac{(\rho V)^n e^{-\rho V}}{n!}. \quad (5)$$

We say that a causal set c is approximated by a spacetime $(\mathcal{M}, g)$ if c can be obtained from $(\mathcal{M}, g_{\mu\nu})$ via a high probability **Poisson sprinkling**. It is obvious that a “regular” lattice cannot do the job since it is not regular in all frames or coordinate systems.

Knowing this, we can now state the fundamental conjecture of CST :

The Hauptvermutung of CST :

The causal set c can be faithfully embedded at density ρ into two distinct spacetimes, $(\mathcal{M}, g)$ and $(\mathcal{M}', g')$ if and only if they are approximately isometric. In other words, if :

$$(c, \prec) + \varepsilon \xrightleftharpoons[\text{Embedding}]{\text{Sprinkling}} (\mathcal{M}, g) \quad \& \quad (c, \prec) + \varepsilon \xrightleftharpoons[\text{Embedding}]{\text{Sprinkling}} (\mathcal{M}', g'), \quad (6)$$

then $(\mathcal{M}, g)$ and $(\mathcal{M}', g')$ differ only at scale smaller than ρ .

As a Quantum Gravity Theory, the Causal Set Theory have the potential to explain the geometrical structure of spacetime in a well-defined manner and also to cure diseases associated with infinitesimal distances :

- ✓ CST cures divergences in Quantum Field Theory : $\lim_{X \rightarrow Y} \langle \phi(X) \phi(Y) \rangle < \infty$.
- ✓ CST cures infinite entanglement entropy of black holes : no *arbitrary* cut-off.
- ✓ CST cures curvature singularity in General Relativity.
- ✓ CST measures metric at sub-Planckian scale [6].
- ✓ CST naturally predicts a cosmological constant of the right magnitude.
- ✓ CST is compatible with Lorentz invariance.
- ✓ Causal sets are maximally discrete, we just need to count
- ✓ Causal sets carry their own metric information.

What are the be-ables¹ and their dynamics in the paradigm of CST ? This report will present a synthesis of the different works that have been carried out as a complete reasoning.

¹Since quantum gravity deal with phenomenon that take place inside black holes or during Big Bang, we will prefer to talk about *be-ables* than observables.

2 Dynamics

A causal set being the “history” of the theory in the sense of “sum-over-histories”, as one approaches the continuum approximation of the theory, it is possible that the effective quantum dynamics begins to resemble the continuum path integral. Given a sample space Ω , the quantum partition function in CST is

$$Z_\Omega \equiv \sum_{c \in \Omega} e^{i \frac{S(c)}{\hbar}}. \quad (7)$$

The standard action for causal set is the Benincasa-Dowker action defined as :

$$S(c_n^i) = \frac{4}{\sqrt{6}} \left[n - N_0^{(n,i)} + 9N_1^{(n,i)} - 16N_2^{(n,i)} + 8N_3^{(n,i)} \right]. \quad (8)$$

Where $N_k^{(n,i)}$ is the total number of k -element order interval in the causal set c_n^i . It can be shown that (8) gives the Einstein-Hilbert action in the continuum limit [7]. The dynamical law will be a rule which assigns probabilities to suitable classes of causal sets. In order to define a discrete bottom-up dynamics for causal sets, we may give, for each n -element causal set c_n , the transition probability from it to each of its possible children. One can then use this rule – technically a probability measure – to demand physically meaningful questions of the theory, this will be the topic of our next discussions.

2.1 Sequential Growth

The Sequential Growth class of stochastic dynamics is a bottom-up approach in which causal sets are grown element by element. By following this procedure, all causal set are constructed. We will detail this formalism by studying successively every physical constrains of the dynamics.

- Internal temporality :

Internal temporality imposes that to respect causality, every new element is born either to the future of, or unrelated to the other elements. In that sense, causal sets evolve only to the future. To clarify this, we will first focus on the two first steps of sequential growth represented by the figure 4.

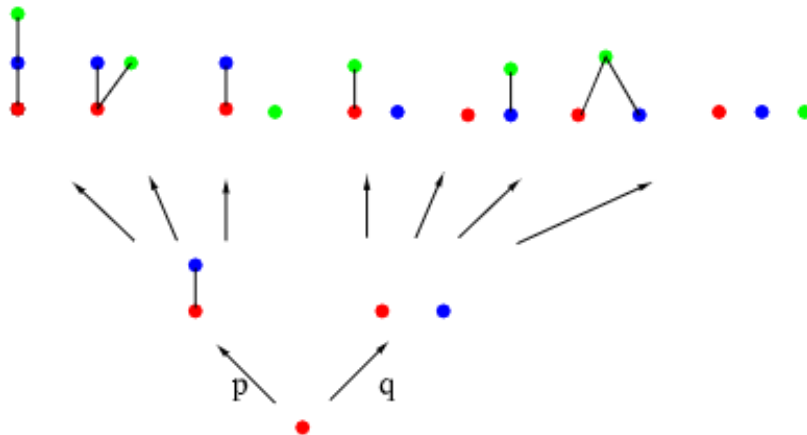


Figure 4: Set of labelled causal set [4].

Let's start with the initial causal set $\tilde{c}_1^1$ which contains a single "red" element e_1 . The tilde sign mean that we are dealing with a labelled "colourful" causal set. The first step occurs, and a new "blue" element e_2 is added either related to (time-like to) e_1 to form $\tilde{c}_2^1$ with probability p , or unrelated to (space-like to) e_1 to form $\tilde{c}_2^2$ with probability $q = 1 - p$. At the second stage, the "green" element e_3 can be added in three different ways if $e_1 \prec e_2$ at the first stage. Only three because, thanks to transitivity, $e_2 \prec e_3$ imposes $e_1 \prec e_3$. If $e_1 \not\prec e_2$, e_3 can be added in four different ways.

- Discrete general covariance :

The “external time” during which the causal set grows is not meant to carry any physical information. In other words, the order of birth reflects a notion of pure gauge. We could relabel without changing the physics. The physical observables are thus constructed from unlabelled causal sets. In figure 4, we remark that the three centred 3-elements causal sets which are made of a chain of two elements and one disconnected element (sometime called “L” causal sets) are different labelling (or colouring) of a same unlabelled “black” causal set. If we group all order-isomorphic causal set together, we get the figure 5 (where all elements with the “L” structure form the element labelled D in 5).

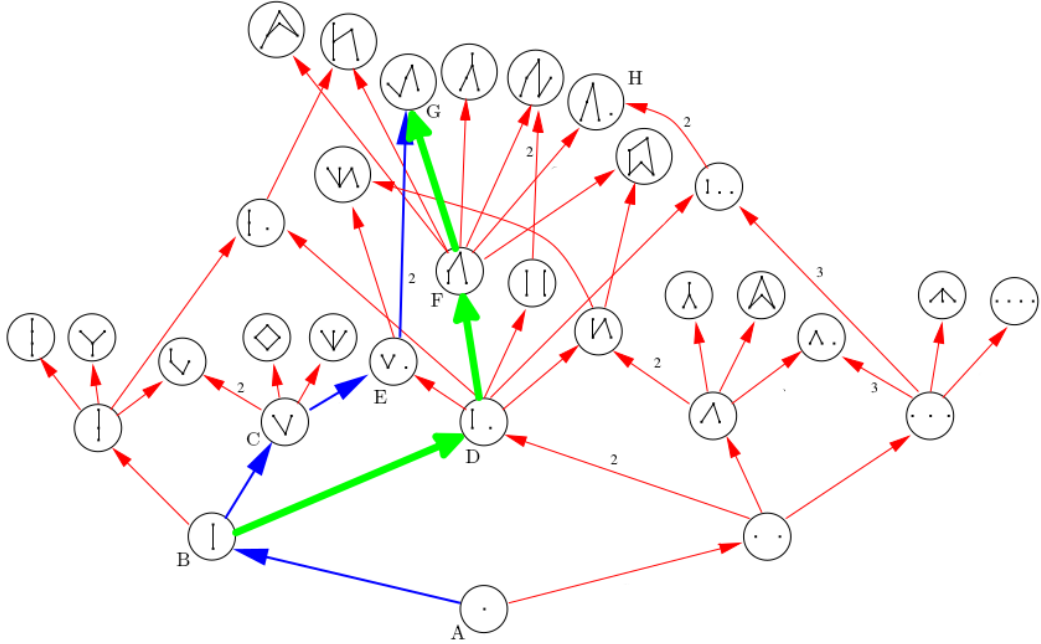


Figure 5: Set of unlabelled causal set [1].

Discrete general covariance is the fact that the dynamic is independent of the labellings. This implies that all path to go from one causal set to another of higher cardinality have the same probability. For example, in the figure 5, the green path $BDFG$, written $B \rightarrow G$, and the blue path $BCEG$, written $B \rightarrow G$, both connect B to G so we have :

$$P(B \rightarrow G) = P(B \rightarrow G) \Leftrightarrow \alpha_{B \rightarrow D} \alpha_{D \rightarrow F} \alpha_{F \rightarrow G} = \alpha_{B \rightarrow C} \alpha_{C \rightarrow E} \alpha_{E \rightarrow G} \quad (9)$$

Where the α symbol denotes the transition probability to go from the parent causal set to a specific child causal set. One of the purpose of this work is to study probabilities like $P(B \rightarrow G)$ or $P(G)$. In the classical case, we will find that $P(B \rightarrow G) = 3P(B \rightarrowtail G)$ because there are three paths that connect B to G. Similarly, $P(G) = P(A \rightarrow G) = 7P(A \rightarrowtail G)$. Difficulties occur when we want to take quantum interferences into account, we need the quantum sequential growth that we are going to detail later for that.

- Bell causality

Birth occurring in some part of a causal set c should be influenced only by the portion of c lying to their past. The causality condition suggests that the probability for a transition should not depend on the spectator set, but this is too strong, and we can see a problem by looking at figure 5 : if the transition depends only on the precursor set then $\alpha_{A \rightarrow B} = \alpha_{F \rightarrow G} = \alpha_{F \rightarrow H}$ and this is absurd if we take $\alpha_{A \rightarrow B} > 1/2$. The Bell Causality condition imposes for $n' < n$:

$$\frac{\alpha_n(c_n^i \rightarrow c_{n+1}^a)}{\alpha_n(c_n^i \rightarrow c_{n+1}^b)} = \frac{\alpha(c_{n'}^i \rightarrow c_{n'+1}^a)}{\alpha(c_{n'}^i \rightarrow c_{n'+1}^b)}. \quad (10)$$

Where the union of the precursor set of the transition $c_{n'}^i \rightarrow c_{n'+1}^a$ and $c_{n'}^i \rightarrow c_{n'+1}^b$ is included in the union of the precursor set of the transition $c_n^i \rightarrow c_{n+1}^a$ and $c_n^i \rightarrow c_{n+1}^b$.

Example :

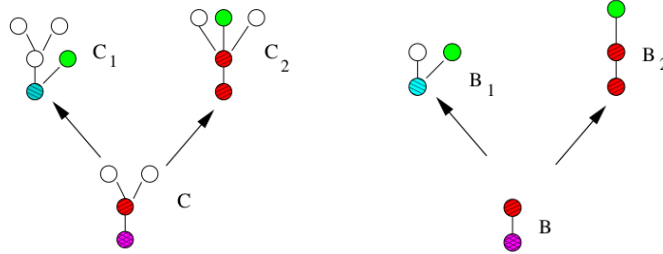


Figure 6: Bell causality - Spectators do not affect relative probability [1].

For the four transitions of figure 2.1, the transition probability satisfy this equality :

$$\frac{P(C \rightarrow C_1)}{P(C \rightarrow C_2)} = \frac{P(B \rightarrow B_1)}{P(B \rightarrow B_2)} \quad (11)$$

- Markov sum rule

Dealing with Markov processes, the sum of every full set of transition probabilities should be 1.

$$\sum_j \alpha(c_n^i \rightarrow c_{n+1}^{j(i)}) = 1 \quad (12)$$

where $\{c_{n+1}^{j(i)}\}$ is the set of children of the causal set c_n^i for the stage n transition.

This set of constraints rules the sequential growth process and thus the global set of all causal sets . By knowing the value of every transition probability, we have access to the overall probability of every causal set possible, and thus we obtain the dynamics. As $n \rightarrow \infty$, this growth process generates the sample space $\tilde{\Omega}$ of countable *labelled* past finite causal sets. Here, another issue occurs : to construct covariant be-ables, we should have unlabelled object that are equivalent classes of a certain number of labelled causal set. But what to do if this number is infinity ? For instance, how to compute the probability of having a 2-chain in a causal set knowing that any labelled causal set that have a 2-chain could be born arbitrarily far in the future. We will use tools from measure theory to deal with this issue.

2.2 General measure space

Let Ω be a non-empty set. $\mathfrak{A} \subseteq \mathcal{P}(\Omega)$ (power set of Ω) is an *algebra* if and only if :

$$(i) : \Omega \in \mathfrak{A}, \quad (ii) : A \in \mathfrak{A} \implies A^c \in \mathfrak{A}, \quad (iii) : A_1, A_2, \dots, A_n \in \mathfrak{A} \implies \bigcup_{k=1}^n A_k \in \mathfrak{A}. \quad (13)$$

The elements $A \in \mathfrak{A}$ are called *events* or *measurable sets*, and the algebra $\mathfrak{A}$ is said to be closed under **finite** unions of event. $(\Omega, \mathfrak{A})$ is a *measurable space*. For finite size causal sets, if $\sim$ denotes equivalence under relabelling, this is enough to define a *covariant set* η such that :

$$\eta \equiv \{\tilde{c} | \tilde{c}' \sim \tilde{c} \implies \tilde{c}' \in \eta\} \subseteq \tilde{\Omega}. \quad (14)$$

Meaning that if a causal set is included in η so does all its relabelling. If we are working with infinite size causal sets, we may want to use a countable union of events. To this purpose, we will need to extend our analysis to the σ -algebra $\mathfrak{S} \subseteq \mathcal{P}(\Omega)$ defined by :

$$(i) : \Omega \in \mathfrak{S}, \quad (ii) : A \in \mathfrak{S} \implies A^c \in \mathfrak{S}, \quad (iii) : A_1, A_2, \dots \in \mathfrak{S} \implies \bigcup_{k=1}^{\infty} A_k \in \mathfrak{S}. \quad (15)$$

$\mathfrak{A}$ is a σ -*algebra* if it is an algebra which is also closed under **countable** unions. The smallest algebra that includes events generated by countable set operations on $\mathfrak{A}$ is its associated σ -algebra $\mathfrak{S}_{\mathfrak{A}}$. Equivalently, one can build covariant events from $\mathfrak{S}_{\mathfrak{A}}$ by taking equivalence classes of causal sets under relabelling. The σ -algebra of covariant events is the quotient σ -algebra $\mathfrak{S}_{\mathfrak{A}}/\sim$.

A *measure* on a measurable space is a map satisfying :

$$(i) : \mu(\emptyset) = 0, \quad (ii) : A_1, \dots, A_n \in \mathfrak{A} \text{ pairwise disjoint} \implies \mu\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n \mu(A_k). \quad (16)$$

The Causal Sets Theory dynamics are given by the *measure space* $(\Omega, \mathfrak{A}, \mu)$ that can be extended to $(\Omega, \mathfrak{S}_{\mathfrak{A}}, \bar{\mu})$ where $\bar{\mu}$ is a countably additive measure. If μ is a scalar measure, the condition for extension is that it must be already countably additive in $\mathfrak{A}$. For vector measures, we need the following simplified theorem.

Simplified Caratheodary-Hahn-Kluvnek (CHK) theorem :

Let $\mathfrak{A}$ be an algebra and $\mathfrak{S}_{\mathfrak{A}}$ the σ -algebra generated by $\mathfrak{A}$. If μ is bounded weakly countably additive vector measure over $\mathfrak{A}$, then :

$$\mu \text{ is strongly additive} \Leftrightarrow \exists! \text{ countably additive extension of } \mu \text{ to } \mathfrak{S}_{\mathfrak{A}}. \quad (17)$$

The terminology used in this theorem is defined in appendix. The full form of this theorem said that four propositions are equivalent, here we look at the equivalence of the two propositions that are of interest. The article [2] prove that : μ of bounded variation $\implies \mu$ strongly additive $\implies \mu$ bounded. Then, the previous simplified theorem could be even more simplified and yields now :

If μ is a weakly countably additive vector measure over $\mathfrak{A}$:

$$\mu \text{ of bounded variation} \Rightarrow \exists! \text{ countably additive extension of } \mu \text{ to } \mathfrak{S}_{\mathfrak{A}}. \quad (18)$$

In order to distinguish between a measure on an algebra and that on a σ -algebra, we could refer to the former as a pre-measure and the latter as a measure (thus, a pre-measure is a measure if the algebra on which it is defined is an algebra).

3 Classical sequential growth

Why are we interested in a classical dynamics for causal sets, when our ultimate aim is a quantum theory of gravity ? One obvious reason is that the classical case, being much simpler, can help us to get used to a relatively unfamiliar type of dynamical formulation.

3.1 Transitive percolation

It is an especially simple instance of a sequential growth dynamics, in which each new element forges a causal bond independently with each existing element with probability p , where $p \in [0, 1]$ is a fixed parameter of the model.

3.1.1 Systematic construction

We can construct a causal set from scratch by the following algorithm :

1. Start with n elements labelled $e_1, e_2, \dots, e_n$ ($n = \infty$ not excluded).
2. With a fixed probability p , introduce a relation between every pair e_i, e_j with $i < j$.
3. Form the transitive closure of these relations (e.g. if $e_2 \prec e_5$ and $e_5 \prec e_8$ then enforce that $e_2 \prec e_8$).

This algorithm could be used for sequential growth by saying that at stage n we start with a n -element causal set c_n^i and a new element e_{n+1} . Then for $i = 1, \dots, n$ we randomly chose if $e_i \prec e_{n+1}$ or not. According to this, the transition probability α_n from c_n^i of size n to a specified child $c_{n+1}^{j(i)}$ of size $n + 1$ is given by :

$$\alpha_n^{(S)} = \alpha_n^{(m, \varpi)} = p^m (1 - p)^{n - \varpi}. \quad (19)$$

Where S is the past of the new element which is born during the transition, $m = m(S)$ the number of maximal elements in S , and $\varpi = |S|$ the cardinality of S . From this, we find that the probability to arrive at the labelled causal set $\tilde{c}_n^i$ is given by :

$$P(\tilde{c}_n^i) = p^L (1 - p)^{\binom{n}{2} - R} \quad (20)$$

where L is the number of links in $\tilde{c}_n^i$ and R the number of relations (or chains). So $\binom{n}{2} - R$ is the number of pair of elements that are unconnected.

3.1.2 Physical requirement for transitive percolation

✓ Internal temporality

Build into our definition of the growth process.

✓ Discrete general covariance

The net probability of a given c_n^i is $P(c_n^i) = W(c_n^i) p^L q^{\binom{n}{2} - R}$ where $W(c_n^i)$ is the number of (natural) labellings of c_n^i , this depends only on the number of link and not on the labellings so the covariance condition is respected.

✓ Bell causality

Consider two different children, one with $(m, \varpi) = (m_1, \varpi_1)$ and the other with $(m, \varpi) = (m_2, \varpi_2)$. Bell Causality imposes that

$$\frac{\alpha_n^{(m_1, \varpi_1)}}{\alpha_n^{(m_2, \varpi_2)}} = \frac{\alpha_{n'}^{(m_1, \varpi_1)}}{\alpha_{n'}^{(m_2, \varpi_2)}} \Leftrightarrow \frac{p^{m_1} q^{n - \varpi_1}}{p^{m_2} q^{n - \varpi_2}} = \frac{p^{m_1} q^{n' - \varpi_1}}{p^{m_2} q^{n' - \varpi_2}} \quad (21)$$

where $n' \leq n$ is the cardinality of the union of the precursor sets of the two transitions.

✓ Markov sum rule

Trivial in a well-defined probabilistic procedure.

3.2 General transition probability

We seek to derive a general prescription which gives, the transition probability from a parent causal set c_n^i to one specified child $c_{n+1}^{j(i)}$. To avoid having to deal with special cases, we will assume throughout that no transition probability vanishes.

3.2.1 Construction of the transitions probabilities

The three following lemmas are proved in appendix :

Lemma 1 : *There is only one free parameter per family.*

Lemma 2 : *The probability to add a completely disconnected element (the “gregarious child transition”) depends only on the cardinality of the parent causal set.*

Lemma 3 : *Each transition probability α_n of stage n has the following form, where the $\lambda_i^{(S)}$ are integers depending on the individual transition in question :*

$$\alpha_n^{(S)} = q_n \sum_{i=0}^n \frac{\lambda_i^{(S)}}{q_i}. \quad (22)$$

The $\lambda_i^{(S)}$ are integers independent of the q_n 's which are the parameters of the growth and probabilities to add a completely disconnected element at stage n , those only depend on the transition in itself. We can find their value by finding the parameter in the transitive percolation case where $q_n = q^n$.

$$\begin{aligned} \alpha_n^{(m, \varpi)} &= p^m (1-p)^{n-\varpi} = (1-q)^m q^{n-\varpi} = \sum_{k=0}^m \binom{m}{k} (-q)^k q^{n-\varpi} = \sum_{k=0}^m \binom{m}{k} (-1)^k \frac{q_n}{q^{\varpi-k}} \\ &= \sum_{k=0}^m \binom{m}{m-k} (-1)^{m-k} \frac{q_n}{q^{\varpi-m+k}} = \sum_{k=\varpi-m}^m \binom{m}{m-(k-\varpi+m)} (-1)^{m-(k-\varpi+m)} \frac{q_n}{q^{\varpi-m+(k-\varpi+m)}} \\ &= \sum_{k=\varpi-m}^m \binom{m}{\varpi-k} (-1)^{\varpi-k} \frac{q_n}{q_k} \stackrel{!}{=} \sum_{i=0}^n \lambda_i^{(m, \varpi)} \frac{q_n}{q_i} \implies \lambda_i^{(m, \varpi)} = \begin{cases} \binom{m}{\varpi-i} (-1)^{\varpi-i} & \text{if } \varpi-m \leq i \leq \varpi \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

At the end, we get for every transition of precursor set S :

$$\alpha_n^{(S)} = \alpha_n^{(m, \varpi)} = \sum_{k=0}^m (-1)^k \binom{m}{k} \frac{q_n}{q^{\varpi-k}} = \frac{\sum_{l=m}^{\varpi} \binom{\varpi-m}{\varpi-l} t_l}{\sum_{j=0}^n \binom{n}{j} t_j} \quad (23)$$

where we introduced the alternative parameters $t_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} \frac{1}{q_k}$ by using an identity :

$$\sum_{k=0}^n \binom{n}{k} (-1)^{n-k} \binom{k}{m} = \delta_m^n \implies \sum_{k=0}^n \binom{n}{k} t_k = \sum_{k=0}^n \sum_{i=0}^k \binom{n}{k} (-1)^{k-i} \binom{k}{i} \frac{1}{q_i} = \sum_{i=0}^n \delta_i^n (-1)^{n-i} \frac{1}{q_i} = \frac{1}{q_n}. \quad (24)$$

3.2.2 Physical requirements for general transition probability

✓ Internal temporality

Build into our definition of the growth process.

✓ Discrete general covariance

The net probability of a given c_n^i is $P(c_n^i) = W f(\{\varpi_i, m_i\})$ where W is the number of (natural) labellings of c_n^i . The probability of a labelled causal set $\tilde{c}_n^i$ is $P(\tilde{c}_n^i) = \prod_{i=0}^{N-1} \alpha(i, \varpi_i, m_i) = f(\{\varpi_i, m_i\})$ with $N = |\tilde{c}_n^i|$. This is a product over all elements $x \in c_n^i$ of invariant quantities that depends only on the structure of $\text{Past}(x)$, this depends only on the number of link and not on the labellings, so the covariance principle is respected.

✓ Bell causality

By considering two different children, one with $(m, \varpi) = (m_1, \varpi_1)$ and the other with $(m, \varpi) = (m_2, \varpi_2)$, where $n' \leq n$ is the cardinality of the union of the precursor sets of the two transitions that leads to those children, Bell Causality implies that :

$$\frac{\alpha_n^{(m_1, \varpi_1)}}{\alpha_n^{(m_2, \varpi_2)}} = \frac{\alpha_{n'}^{(m_1, \varpi_1)}}{\alpha_{n'}^{(m_2, \varpi_2)}} \Leftrightarrow \frac{\sum_{k=0}^{m_1} (-1)^k \binom{m_1}{k} \frac{q_n}{q_{\varpi_1-k}}}{\sum_{k=0}^{m_2} (-1)^k \binom{m_2}{k} \frac{q_n}{q_{\varpi_2-k}}} = \frac{\sum_{k=0}^{m_1} (-1)^k \binom{m_1}{k} \frac{q_{n'}}{q_{\varpi_1-k}}}{\sum_{k=0}^{m_2} (-1)^k \binom{m_2}{k} \frac{q_{n'}}{q_{\varpi_2-k}}} \quad (25)$$

$$\Leftrightarrow \frac{\sum_{k=0}^{m_1} (-1)^k \binom{m_1}{k} \frac{1}{q_{\varpi_1-k}}}{\sum_{k=0}^{m_2} (-1)^k \binom{m_2}{k} \frac{1}{q_{\varpi_2-k}}} = \frac{\sum_{k=0}^{m_1} (-1)^k \binom{m_1}{k} \frac{1}{q_{\varpi_1-k}}}{\sum_{k=0}^{m_2} (-1)^k \binom{m_2}{k} \frac{1}{q_{\varpi_2-k}}} \quad (26)$$

The ratios depend only on precursor set structure, so Bell Causality is always respected.

✓ Markov sum rule

This constraint imposes an infinite set of identities which must hold in our dynamics :

$$\sum_{i=0}^{N-1} \alpha(i, \varpi_i, m_i) = 1 \Leftrightarrow \sum_S \sum_l t_l \binom{|S| - m(S)}{l - m(S)} = \sum_j t_j \binom{n}{j} \Leftrightarrow \forall l, \sum_S \binom{|S| - m(S)}{l - m(S)} = \binom{n}{l}. \quad (27)$$

3.2.3 Corresponding measure space

In the classical growth models, the probability measure space is $(\tilde{\Omega}, \mathfrak{A}, \mu)$ where $\tilde{\Omega}$ is the space of all past finite labelled causal sets, $\mathfrak{A}$ is the event algebra closed under finite set operations over $\tilde{\Omega}$ and μ is the probability measure. The dynamic of the causal sets is a specification of the measure over $\mathfrak{A}$. To find this, we define the *cylinder set* as follows :

$$\text{cyl}(\tilde{c}_n^i) \equiv \{\tilde{c} \in \tilde{\Omega} | \tilde{c}|_n = \tilde{c}_n^i\} = \bigcup_{j(i)} \text{cyl}(\tilde{c}_{n+1}^{j(i)}) \subset \tilde{\Omega} \quad (28)$$

where $\tilde{c}|_n$ denotes the first n elements of the labelled causal set $\tilde{c} \in \tilde{\Omega}$. The cylinder sets satisfy the so called nesting property for $m > n$:

$$\text{cyl}(\tilde{c}_m^i) \cap \text{cyl}(\tilde{c}_n^j) \neq 0 \implies \text{cyl}(\tilde{c}_m^i) \subset \text{cyl}(\tilde{c}_n^j). \quad (29)$$

The event algebra $\mathfrak{A}$ is generated by taking finite unions, intersections and complements of cylinder sets. The nesting property (29) implies that for any $A \in \mathfrak{A}$, there exists a smallest integer $n_A < \infty$ and a subset $S_A \subset \{1, \dots, |\Omega_{n_A}|\}$ such that $A = \bigcup_{k \in S_A} \text{cyl}(\tilde{c}_{n_A}^k)$. The cylinder sets are subsets of Ω and thus are element of $\mathfrak{A}$, the measure act on it as :

$$\mu(\text{cyl}(\tilde{c}_n^i)) \equiv P(\tilde{c}_n^i) \quad (30)$$

and so thanks to (16),

$$\mu(A) = \sum_{k \in S_A} P(\tilde{c}_{n_A}^k). \quad (31)$$

In the classical case we deal with scalar measure and thus the extension to a σ -algebra is guaranteed : the necessary condition is just the countable additivity of the pre-measure. For the quantum case, we are going to study vector measure, where the extension is not trivial.

4 Quantum sequential growth

Previously, we constructed causal set dynamics in classical sequential growth. It is stochastic only in the purely classical sense, lacking quantum interference. One of the most important open problem is the field of Causal Set Theory is the construction from first principles of a viable quantum dynamics for causal sets [4]. The rest of this report is dedicated to quantum sequential growth and to the first building blocks of the construction of a quantum measure.

4.1 Decoherence functional

Until now, in the classical case if we defined an event as a set of histories or disjointed elementary event, $E = \{\gamma_1, \gamma_2, \dots, \gamma_N\}$, the pre-measure comes as :

$$\mu(E) = \mu\left(\bigcup_{i=1} \gamma_i\right) = \sum_{i=1} \mu(\gamma_i). \quad (32)$$

For quantum probabilities, interferences are added as follows [8] :

$$\mu(E) = \mu\left(\bigcup_{i=1} \gamma_i\right) = \sum_{i=1} \mu(\gamma_i) + \sum_{\substack{i,j=1 \\ i < j}} I(\gamma_i, \gamma_j) \quad (33)$$

where the $I(x, y)$ is *interference term* between events x and y and μ is now the quantum pre-measure. Because of interferences, the initial finite additive of the pre-measure (32) must not be always valid anymore. For pairwise disjoint $A, B, C \in \mathfrak{A}$, the sum rule is replaced by the quantum sum rule :

$$\mu(A \cup B \cup C) = \mu(A \cup B) + \mu(A \cup C) + \mu(B \cup C) - \mu(A) - \mu(B) - \mu(C) \quad (34)$$

Proof :

$$\begin{aligned} \mu(A \cup B \cup C) &= \mu(A) + \mu(B) + \mu(C) + I(A, B) + I(B, C) + I(A, C) \\ &= \mu(A) + \mu(B) + \mu(C) + (\mu(A \cup B) - \mu(A) - \mu(B)) \\ &\quad + (\mu(B \cup C) - \mu(B) - \mu(C)) + (\mu(A \cup C) - \mu(A) - \mu(C)) \\ &= \mu(A \cup B) + \mu(A \cup C) + \mu(B \cup C) - \mu(A) - \mu(B) - \mu(C). \quad \square \end{aligned}$$

Thus, by extending our classical measure μ in the quantum world, it loses additivity. We need to construct another additive (pre-)measure that we will call vector (pre-)measure μ_v . To do this we first define the *decoherence functional* $D : \mathfrak{A} \times \mathfrak{A} \rightarrow \mathbb{C}$ which respects :

- *Hermiticity* : $\forall A, B \in \mathfrak{A}, D(A, B) = D(B, A)^*$.
- *Bi-additivity* : $\forall A, B, C \in \mathfrak{A} \text{ — } B \cap C = \emptyset,$

$$D(A, B \cup C) = D(A, B) + D(A, C) \text{ \& } D(B \cup C, A) = D(B, A) + D(C, A) \quad (35)$$

- *Normalisation* : $D(\Omega, \Omega) = 1$.
- *Strong positivity* : for any $\{A_i\}$ finite collection in $\mathfrak{A}$, $M_{ij} = D(A_i, A_j)$ has non-negative eigenvalue.

Our initial formalism could now be written with the decoherence functional like this :

$$\text{for } x, y \in \mathfrak{A}, x \cap y = \emptyset : \quad \mu(x) = D(x, x) \quad I(x, y) = D(x, y) + D(y, x) = 2\text{Re}(D(x, y)). \quad (36)$$

Our goal will be to construct a quantum vector measure out of it. Let's first construct a quantum vector pre-measure to recover finite additivity.

4.2 Quantum vector pre-measure

To be quantum, our pre-measure must be $\mu_v : \mathfrak{A} \rightarrow \mathcal{H}$, where $\mathcal{H}$ is the appropriate Hilbert space we need to find.

4.2.1 Construction of the inner product vector space $(\mathcal{H}_1, +, \cdot, \langle \cdot, \cdot \rangle_1)$

Let's define $\mathcal{H}_1$ as set of all complex-valued functions on $\mathfrak{A}$ which are non-zero only on a finite number of events. The three functions $+$, $\cdot$ and the "inner product" of the set $(\mathcal{H}_1, +, \cdot, \langle \cdot, \cdot \rangle_1)$ are defined as follows :

$$\forall u, v \in \mathcal{H}_1, A \in \mathfrak{A}, \quad (u + v)(A) \equiv u(A) + v(A). \quad (37)$$

$$\forall u \in \mathcal{H}_1, A \in \mathfrak{A}, \lambda \in \mathbb{C}, \quad (\lambda \cdot u)(A) \equiv \lambda u(A). \quad (38)$$

$$\forall u, v \in \mathcal{H}_1, \quad \langle u, v \rangle_1 \equiv \sum_{A \in \mathfrak{A}} \sum_{B \in \mathfrak{A}} u^*(A) v(B) D(A, B). \quad (39)$$

Here, the "inner product" is degenerate since if we take the non-zero vector $w \in \mathcal{H}_1$ defined as [9] :

$$w(x) \equiv \begin{cases} 1 & \text{if } x = A \\ 1 & \text{if } x = B \\ -1 & \text{if } x = A \cup B \\ 0 & \text{otherwise} \end{cases} \quad (40)$$

for two non-empty, disjoint events $A, B \in \mathfrak{A}$ we can show that $\|w\|_1 = 0$.

Proof :

$$\begin{aligned} \|w\|_1 = \langle w | w \rangle &= \sum_{X, Y \in \mathfrak{A}} w(X)^* w(Y) D(X, Y) = D(A, A) + D(A, B) - D(A, A \cup B) + D(B, B) \\ &+ D(B, A) - D(B, A \cup B) + D(A \cup B, A \cup B) - D(A \cup B, A) - D(A \cup B, B) = 0. \quad \square \end{aligned}$$

So $\langle x, x \rangle = 0 \nRightarrow x = 0$, then $\langle \cdot, \cdot \rangle_1$ is not a true inner-product, and so $(\mathcal{H}_1, +, \cdot, \langle \cdot, \cdot \rangle_1)$ is not yet a Hilbert space. Another step is needed to construct a well-defined Hilbert space out of it.

4.2.2 Construction of the Hilbert space of histories $(\mathcal{H}_2, +, \cdot, \langle \cdot, \cdot \rangle_2)$

We define the equivalence class $\sim$ between two Cauchy sequences $\{u_n\}$ and $\{v_n\}$ as :

$$\{u_n\} \sim \{v_n\} \Leftrightarrow \lim_{n \rightarrow +\infty} \|u_n - v_n\|_1 = 0. \quad (41)$$

By defining $\mathcal{H}_2 \equiv \mathcal{H}_1 / \sim$, we can show that $(\mathcal{H}_2, +, \cdot, \langle \cdot, \cdot \rangle_2)$ is a Hilbert space. With the $\sim$ -equivalence class of a Cauchy sequence $\{u_n\}$ denoted by $[u_n]$, the three functions $+$, $\cdot$ and the true inner product of the set $(\mathcal{H}_2, +, \cdot, \langle \cdot, \cdot \rangle_2)$ are defined as follows :

$$\forall [u_n], [v_n] \in \mathcal{H}_2, \quad [u_n] + [v_n] \equiv [u_n + v_n]. \quad (42)$$

$$\forall [u_n] \in \mathcal{H}_2, \lambda \in \mathbb{C}, \quad \lambda \cdot [u_n] \equiv [\lambda u_n]. \quad (43)$$

$$\forall [u_n], [v_n] \in \mathcal{H}_2, \quad \langle [u_n], [v_n] \rangle_2 \equiv \lim_{n \rightarrow +\infty} \langle u_n, v_n \rangle_1. \quad (44)$$

The inner product is no longer degenerate, $\mathcal{H}_2$ is a Hilbert space.

4.2.3 Construction of the vector pre-measure

We define the quantum vector pre-measure to be $\mu_v(A) = [\chi_A]$ where $\{\chi_A\}$ is the constant Cauchy sequence of the indicator function :

$$\chi_A(B) = \begin{cases} 1 & \text{if } B = A \\ 0 & \text{if } B \neq A \end{cases}.$$

Thus,

$$\langle \mu_v(A), \mu_v(B) \rangle_2 = \langle \chi_A, \chi_B \rangle_1 = \sum_{A' \in \mathfrak{A}} \sum_{B' \in \mathfrak{A}} \chi_A^*(A') \chi_B(B') D(A', B') = D(A, B). \quad (45)$$

For any disjoint pair $A, B \in \mathfrak{A}$,

$$\begin{aligned} \|\chi_{A \cup B} - \chi_A - \chi_B\|_1 &= \sqrt{\langle \chi_{A \cup B} - \chi_A - \chi_B, \chi_{A \cup B} - \chi_A - \chi_B \rangle} \\ &= \sqrt{\sum_{A', B' \in \mathfrak{A}} (\chi_{A \cup B}(A') - \chi_A(A') - \chi_B(A')) (\chi_{A \cup B}(B') - \chi_A(B') - \chi_B(B')) D(A', B')} \\ &= 0 \text{ because } A \cap B = \emptyset \implies \forall x \in \mathfrak{A}, \chi_{A \cup B}(x) = \chi_A(x) + \chi_B(x). \end{aligned}$$

This implies that μ_v is finitely additive in $\mathcal{H}_2$ because $\mu_v(A \cup B)$ and $\mu_v(A) + \mu_v(B)$ belong to the same equivalence class. In the following, we will use kets :

$$\forall A, B \in \mathfrak{A}, |A\rangle \equiv \mu_v(A) \text{ \& } \langle A|B\rangle \equiv D(A, B). \quad (46)$$

The challenge is now to extend μ_v to the σ -algebra to have a complete quantum vector measure.

4.3 Complex sequential growth CSG

The first step toward an extended quantum vector measure is to study the complex version of classical sequential growth to define a complex vector measure $\mu_v : A \rightarrow \mathbb{C}$. This is the simplest quantum version of the growth models.

4.3.1 Construction of the transitions probabilities

Complex sequential growth is a straightforward generalization of classical sequential growth, the transition amplitudes contain now complex parameters $q_n, t_n \in \mathbb{C}$:

$$\alpha_n^{(S)} = \alpha_n^{(m, \varpi)} = \sum_{k=0}^m (-1)^k \binom{m}{k} \frac{q_n}{q^{\varpi-k}} = \frac{\sum_{l=m}^{\varpi} \binom{\varpi-m}{\varpi-l} t_l}{\sum_{j=0}^n \binom{n}{j} t_j} \in \mathbb{C}. \quad (47)$$

The measure of $\text{cyl}(\tilde{c}_n^i) \in \mathfrak{A}$ yields (with ϕ an arbitrary phase) :

$$|\tilde{c}_n^i\rangle \equiv \mu_v(\text{cyl}(\tilde{c}_n^i)) = e^{i\phi} \prod_m \alpha(\tilde{c}_m \rightarrow \tilde{c}_{m+1}) \in \mathbb{C}. \quad (48)$$

The product is over transition along the unique set of transition that leads from $\tilde{c}_1^1$ to $\tilde{c}_n^i$.

4.3.2 Physical requirement on CSG

✓ Internal temporality

Build into our definition of the growth process.

✓ Discrete general covariance

We still have that whenever $\tilde{c}_n^i$ and $\tilde{c}_n^j$ are order-isomorphic, then $|\tilde{c}_n^i\rangle = |\tilde{c}_n^j\rangle$.

✓ Bell causality :

Same thing than in the classical case,

$$\frac{\alpha_n^{(m_1, \varpi_1)}}{\alpha_n^{(m_2, \varpi_2)}} = \frac{\alpha_{n'}^{(m_1, \varpi_1)}}{\alpha_{n'}^{(m_2, \varpi_2)}} \Leftrightarrow \frac{\sum_{k=0}^{m_1} (-1)^k \binom{m_1}{k} \frac{1}{q^{\varpi_1-k}}}{\sum_{k=0}^{m_2} (-1)^k \binom{m_2}{k} \frac{1}{q^{\varpi_2-k}}} = \frac{\sum_{k=0}^{m_1} (-1)^k \binom{m_1}{k} \frac{1}{q^{\varpi_1-k}}}{\sum_{k=0}^{m_2} (-1)^k \binom{m_2}{k} \frac{1}{q^{\varpi_2-k}}}. \quad (49)$$

✓ Markov sum rule

For every child $\tilde{c}_{n+1}^{j(i)}$ of $\tilde{c}_n^i$ there exists a transition $\alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)})$ such that,

$$|\tilde{c}_{n+1}^{j(i)}\rangle = \alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)}) |\tilde{c}_n^i\rangle. \quad (50)$$

The transition amplitude $\alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)})$ is a special case of transition matrix.

$$\mu_v(\text{cyl}(\tilde{c}_n^i)) = \mu_v\left(\bigcup_{j(i)} \text{cyl}(\tilde{c}_{n+1}^{j(i)})\right) = \sum_{j(i)} \mu_v(\text{cyl}(\tilde{c}_{n+1}^{j(i)})) = \sum_{j(i)} \alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)}) |\tilde{c}_n^i\rangle = |\tilde{c}_n^i\rangle \quad (51)$$

In the quantum case, the Markov sum rule yields :

$$\sum_{j(i)} \alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)}) = 1. \quad (52)$$

4.3.3 Corresponding measure space

Since the transition probabilities are now complex, we have the following triangle inequality :

$$\sum_{j(i)} |\alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)})| \geq \left| \sum_{j(i)} \alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)}) \right| = 1. \quad (53)$$

We can then define $\zeta_n^i \geq 0$ such as $\sum_{j(i)} |\alpha(\tilde{c}_n^i \rightarrow \tilde{c}_{n+1}^{j(i)})| = 1 + \zeta_n^i \geq 1$. And at every stage n , the finiteness of $\tilde{\Omega}_n$ allow us to define :

$$\zeta_n^{\max} \equiv \max_{c_n^i \in \Omega_n} \zeta_n^i, \quad \zeta_n^{\min} \equiv \min_{c_n^i \in \Omega_n} \zeta_n^i. \quad (54)$$

With these quantities, we can write the theorem that give the necessary conditions to extend a complex measure to a σ -algebra :

Surya-Zalel (SZ) theorem :

μ_v is of bounded variation if $\sum_{n=1}^{\infty} \zeta_n^{\max}$ converges μ_v is not of bounded variation if $\sum_{n=1}^{\infty} \zeta_n^{\min}$ diverges.

The proof of this theorem will be developed in appendix.

In the context of Complex Sequential Growth, we may determine $\zeta_n^{\max}$ and $\zeta_n^{\min}$ in terms of the parameters t_i 's, (also done in appendix) :

$$\zeta_n^{\max} = \frac{\sum_{k=0}^n \binom{n}{k} |t_k|}{\left| \sum_{k=0}^n \binom{n}{k} t_k \right|} - 1 = \zeta_n^a, \quad (55)$$

$$\zeta_n^{\min} = \sum_{\varpi=1}^n \frac{\left| \sum_{k=1}^{\varpi-1} \binom{\varpi-1}{k-1} t_k \right|}{\left| \sum_{k=0}^n \binom{n}{k} t_k \right|} + \frac{|t_0|}{\left| \sum_{k=0}^n \binom{n}{k} t_k \right|} - 1 = \zeta_n^c. \quad (56)$$

4.4 Complex percolation $\mathbb{CP}$

Complex percolation is a natural quantum generalization of transitive percolation, in which real probabilities are replaced by complex amplitudes. Thanks to (36) and (46) where kets are complex number, the only decoherence functional for a complex vector measure with $\mathcal{H} = \mathbb{C}$ yields :

$$D(\text{cyl}(\tilde{c}_n^i), \text{cyl}(\tilde{c}_n^j)) = \mu_v(\tilde{c}_n^i)^* \mu_v(\tilde{c}_n^j) \quad (57)$$

As it is a generalization of transitive percolation, the probability amplitude for a new element to be timelike to another is given by the complex parameter p , and the probability amplitude to be space-like is given by q . As in the classical case, we see thanks to (48) that only one parameter is needed :

$$\mu_v(\tilde{c}_1^1) = \mu_v(\tilde{c}_2^1) + \mu_v(\tilde{c}_2^2) = \mu_v(\tilde{c}_1^1)\alpha(\tilde{c}_1^1 \rightarrow \tilde{c}_2^1) + \mu_v(\tilde{c}_1^1)\alpha(\tilde{c}_1^1 \rightarrow \tilde{c}_2^2) = \mu_v(\tilde{c}_1^1)(p + q) \implies p + q = 1. \quad (58)$$

Thanks to the triangle inequality, (36) and (57) we could write :

$$\mu(\tilde{\Omega}) = 1 = |\mu_v(\tilde{\Omega})|^2 \implies |\mu_v(\tilde{\Omega})| = \left| \sum_{i=1}^{|\tilde{\Omega}_n|} \mu_v(\tilde{c}_n^i) \right| = 1 \implies \sum_{i=1}^{|\tilde{\Omega}_n|} |\mu_v(\tilde{c}_n^i)| \geq 1. \quad (59)$$

In the case $n = 2$, since $p + q = 1$:

$$p \in \mathbb{R} \Leftrightarrow |p| + |q| = 1. \quad (60)$$

We now focus on the case $n > 2$. Equality in (59) is equivalent to collinearity of the transition amplitudes :

$$\sum_{i=1}^{|\tilde{\Omega}_n|} |\alpha(\tilde{c}_n^i)| = 1 \Leftrightarrow \forall i, j \leq |\tilde{\Omega}_n|, \alpha(\tilde{c}_n^i) = k_{ij} \alpha(\tilde{c}_n^j) \text{ with } k_{ij} \in \mathbb{R}. \quad (61)$$

Let denote by $\tilde{c}_n^v$ an $n - 2$ -chain topped with a “V”, i.e. with e_n, e_{n-1} to the immediate future of the maximal element e_{n-2} of the $n - 2$ chain $\tilde{c}_{n-2}^c$, and unrelated to each other as described in figure 7.

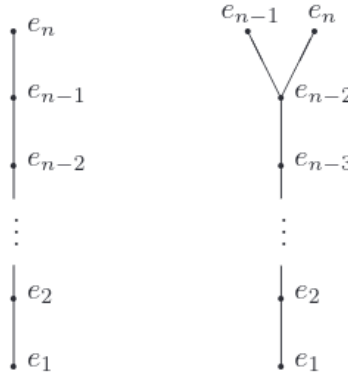


Figure 7: Hasse diagrams for $\tilde{c}_n^c$ and $\tilde{c}_n^v$.

Then we have $\alpha(\tilde{c}_n^c) = p^{n-1}$ and $\alpha(\tilde{c}_n^v) = p^{n-1}q$, so requiring collinearity of these amplitudes is therefore equivalent to requiring that p is real. Then we have $\forall n$:

$$\sum_{i=1}^{|\tilde{\Omega}_n|} |\alpha(\tilde{c}_n^i)| = 1 \Leftrightarrow p \in \mathbb{R}. \quad (62)$$

It is shown in appendix that :

$$p \notin \mathbb{R} \Leftrightarrow \sum_{i=1}^{|\tilde{\Omega}_n|} |\alpha(\tilde{c}_n^i)| > 1 \implies \mu_v \text{ is not of bounded variation.} \quad (63)$$

So this implies that, to extend the measure to the σ -algebra, p must be real. In other words, this extension occur for classical transitive percolation and not for the complex one.

5 Conclusion

In this report, we showed the big picture of what is Causal Set Theory and some aspects that are needed to be worked on in order to reinforce the explanatory potential of this Quantum Gravity paradigm. After defining the objects that take role in CST, we went through the steps to construct the quantum measure that we need to extend. We found the conditions for the extension of a measure in the classical and complex case, we now need to go beyond $\mathbb{C}$ to find the conditions of extension of a Hilbert-space valued quantum measure. Namely, the failure of a quantum measure to have a “mathematical extension” does not mean that it cannot have a “physical extension”, there is room for exploration in this area. I have one month and a half of internship left, my work will now focus on path integrals, random walks, measure theory and helping Sumati Surya for the writing of the first book on Causal Set Theory.

From a personal point of view, this internship allowed me to learn the functioning of a research laboratory in another country. I have discovered new ways for practising, working and teaching physics. The life is extremely different of what I am used to. I also had the opportunity to discover a brand-new subject from scratch, as CST is not popular in France. Even if my PhD project is not directly connected to it, it was very rewarding.

6 Appendix A : Terminology

- **Tangent space** $T_p M$ to the smooth manifold M at the point $p \in M$: vector space of all directional derivatives along curves at p .
- **Pushforward** of a smooth map : Let M and N be smooth manifolds with $p \in M$, and let $\phi : M \rightarrow N$ be a smooth map. The pushforward of ϕ is the map $\phi_* : T_p M \rightarrow T_{\phi(p)} N$ defined by

$$(\phi_* X)f = X(f \circ \phi) \quad (64)$$

for any tangent vector $X \in T_p M$ and any smooth function $f \in C^\infty(N)$.

- Ω : space of all past-finite causal set.
- $\Omega_n \subset \Omega$: space of causal sets of fixed cardinality n .
- $c \in \Omega$: any past-finite causal set.
- $c_n \in \Omega_n$: any past-finite causal set of cardinality n .
- $c_n^i \in \Omega_n, i = 1, \dots, |\Omega_n|$: one specific past-finite causal set.
- **Relation** : Pair of elements (x, y) of a causal set such that $x \prec y$
- **Link** : Relation (x, y) where there is no element z such that $x \prec z \prec y$.
- n element **chain** (or n -chain) $c^c : \forall e_i, e_j \in c_n^c$, either $e_i \prec e_j$ or $e_j \prec e_i$.
- n element **antichain** (or n -antichain) $c^a : \forall e_i, e_j \in c_n^a$, we have $e_i \not\prec e_j$.
- **Past** of a subset $\mathcal{A} \equiv \{a_i\}$ of c : $\text{Past}(\mathcal{A}) \equiv \{e \in c \text{ such that } \forall i, e \prec a_i\}$
- **Future** of a subset $\mathcal{A} \equiv \{a_i\}$ of c : $\text{Fut}(\mathcal{A}) \equiv \{e \in c \text{ such that } \forall i, a_i \prec e\}$
- n element **order interval** : $\mathbf{I}[e_i, e_j] \equiv \text{Fut}(\{e_i\}) \cap \text{Past}(\{e_j\})$ such that $|\mathbf{I}[e_i, e_j]| = n$
- c and c' are **order-isomorphic** : c and c' are different labelling of a same unlabelled causal set.
- **Past-finite** causal set $c : \forall e \in c, |\text{Past}(e)| < \infty$.

- **Natural labelling** : labelling ℓ such that $\forall e, e' \in c_n^i, e \prec e' \implies \ell(e) < \ell(e')$.
- **Family** : Set of causal sets which can be formed by adjoining a single maximal element to a given causet in which :
 - **Parent** : The causet from which they come.
 - **Children** : The rest of the set, two children are **siblings**.
 - **Gregarious child** : The child formed by adjoining an element spacelike to every other element.
 - **Timid child** : The child formed by adjoining an element to the future of every element.
- **Precursor set** of a transition : Past of the new element.
- **Spectator set** of a transition : complementary of the precursor set, in the parent set.
- **Partial stem** : Finite subset of a causal set which contains its own past.
- **Full stem** : Partial stem such that every element of its complement lies to the future of (at least) one of its maximal elements.
- **Maximal element** : element that it is to the past of no other element.
- α_n : an arbitrary transition probability from Ω_n to Ω_{n+1} .
- β_n : a non-timid transition probability from Ω_n to Ω_{n+1} , the precursor set is not the entire parent.
- γ_n : a timid transition probability from Ω_n to Ω_{n+1} , the precursor set is the entire parent.
- q_n : a gregarious transition probability from Ω_n to Ω_{n+1} , a transition probability from the n -antichain to the $(n+1)$ -antichain. Remark : $q_0 \equiv 1$ because the universe exist.
- **Finite partition** π of an event $A \in \mathfrak{A}$: set of disjoint subsets $\{a_i\}_{1 \leq i \leq N}$ of A , such that $\bigcup_{i=1}^N a_i = A$.
- μ is **finitely additive** : for every finite sequence $\{A_1, A_2, \dots, A_N\}$ of pairwise disjoint element of $\mathfrak{A}$,

$$\mu\left(\bigcup_{k=1}^N A_k\right) = \sum_{k=1}^N \mu(A_k).$$

- μ is **countably additive** : for every infinite sequence $\{A_1, A_2, \dots\}$ of pairwise disjoint element of $\mathfrak{S}$,

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

- μ is **weakly countably additive** : $\forall x^* \in \mathcal{H}^*$ (dual space), $x^*(\mu)$ is countably additive.
- μ is **strongly additive** : for every infinite sequence $\{A_1, A_2, \dots\}$ of pairwise disjoint element of $\mathfrak{A}$,

$$\left\| \sum_{n=1}^{\infty} \mu(\alpha_n) \right\| < \infty.$$

- **Variation** $|\mu|$ of a measure μ : $|\mu|(A) \equiv \sup_{\pi} \sum_{a_i \in \pi} |\mu(a_i)|$ where π is a finite partition of $A \in \mathfrak{A}$.
- μ is **of bounded variation** if $|\mu|(\Omega) < \infty$.
- **Semi-variation** $\|\mu\|$ of a measure μ : $\|\mu_v\|(A) = \sup_{x^*} \{|x^*(\mu)|(A); x^* \in \mathcal{H}^*, \|x^*\| \leq 1\} < \infty$.
- μ is **bounded** : $\|\mu\|(\Omega) < \infty$.

7 Appendix B : Proofs of the theorems

7.1 The three lemmas of classical sequential growth [1]

7.1.1 Lemma 1

There is only one free parameter per family.

Consider a parent and its children. For every non-timid child there is a Bell causality equation for the ratio of probability of the transition from the parent to this child over the probability of the gregarious transition of the family :

$$\frac{\beta_n}{q_n} = \frac{\alpha_k}{q_k} \text{ with } k < n. \quad (65)$$

So the ratio $\frac{\beta_n}{q_n}$ of a stage n family depends on the parameters of a lower stage k family. Hence, all the β_n 's are determined by knowing q_n . The timid transition could be determined from the other transitions probability by the Markov sum rule. Thus, knowing q_n is enough to determined all the transitions probabilities of the family. $\square$

7.1.2 Lemma 2

The probability to add a completely disconnected element (the “gregarious child transition”) depends only on the cardinality of the parent causal set.

Consider an arbitrary causal set c_n^i represented by a cross-hatched circle. From it we can construct the causal sets $\mathbf{D}, \mathbf{H} \in \Omega_{n+1}$, $\mathbf{A}, \mathbf{C}, \mathbf{F} \in \Omega_{n+2}$ and $\mathbf{B}, \mathbf{E}$ and $\mathbf{G} \in \Omega_{n+3}$ as represented in figure 8. Lower case letter are for transition probability.

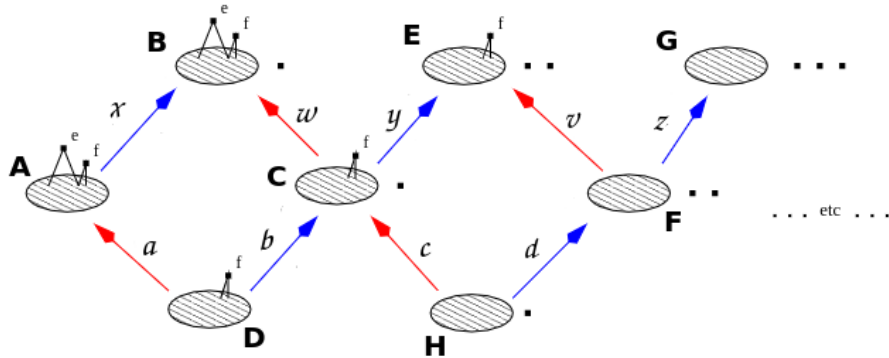


Figure 8: Equality of “gregarious child” transitions [1].

Discrete general covariance imposes

$$ax = bw. \quad (66)$$

Bell causality imposes

$$\frac{y}{w} = \frac{b}{a} \implies ax = bw = ay. \quad (67)$$

So this show easily that : $x = y$. Knowing this we can prove similarly that $y = z$. Continuing in this way until we reach the antichain c_n^c shows finally that $x = q_{n+3}$ depending only on $n + 3$ and so only on the cardinality of the parent causal set. $\square$

7.1.3 Lemma 3

Each transition probability α_n of stage n has the form $\alpha_n = q_n \sum_{i=0}^n \frac{\lambda_i}{q_i}$ where the λ_i are integers depending on the individual transition in question.

Let's check first that this is true for stage 0 :

$$\alpha_0 = q_0 \frac{\lambda_0}{q_0} = \lambda_0 \equiv 1.$$

We assume now that Lemma 3 is true for stage $n - 1$.

Case 1 : Consider a non-timid transition probability β_n of stage n . Bell causality gives :

$$\frac{\beta_n}{q_n} = \frac{\alpha_{n-1}}{q_{n-1}} \implies \beta_n = \alpha_{n-1} \frac{q_n}{q_{n-1}} = \sum_{i=0}^{n-1} \lambda_i \frac{q_{n-1}}{q_i} \frac{q_n}{q_{n-1}} = \sum_{i=0}^{n-1} \lambda_i \frac{q_n}{q_i}.$$

If Lemma 3 is true at stage $n - 1$ then it is true for non-timid transitions at stage n

Case 2 : For a timid transition probability γ_n , we use the Markov sum rule :

$$\gamma_n = 1 - \sum_i \beta_{n,i} = 1 - \sum_j \sum_{i=0}^{n-1} \frac{\lambda_{ji}}{q_i} q_n = 1 + \sum_{i=0}^{n-1} \frac{-\sum_j \lambda_{ji}}{q_i} q_n = 1 + \sum_{i=0}^{n-1} \frac{\lambda_i}{q_i} q_n = \sum_{i=0}^n \lambda_i \frac{q_n}{q_i}$$

by taking $\lambda_i = -\sum_j \lambda_{ji}$ for $i < n$ and $\lambda_n = 1$. If Lemma 3 is true at stage $n - 1$ then it is true for timid transitions at stage n .

In the following, we will omit the tilde $\sim$ to write labelled objects. $\square$

7.2 Surya-Zalel theorem [2]

μ_v is of bounded variation if $\sum_{n=1}^{\infty} \zeta_n^{\max}$ converges μ_v is not of bounded variation if $\sum_{n=1}^{\infty} \zeta_n^{\min}$ diverges

Let's first define

$$S_n \equiv \sum_{i=1}^{|\Omega_n|} || |c_n^i\rangle ||. \quad (68)$$

This implies that at the next stage :

$$S_{n+1} = \sum_{k=1}^{|\Omega_{n+1}|} || |c_{n+1}^k\rangle || = \sum_{i=1}^{|\Omega_n|} \sum_{j(i)} || |c_{n+1}^{j(i)}\rangle || \quad (69)$$

$$= \sum_{i=1}^{|\Omega_n|} \sum_{j(i)} |A(c_n^i \rightarrow c_{n+1}^{j(i)})| || |c_n^i\rangle || = \sum_{i=1}^{|\Omega_n|} (1 + \zeta_n^i) || |c_n^i\rangle ||. \quad (70)$$

And since $\zeta_r^{\min} \leq \zeta_r^i \leq \zeta_r^{\max}$ we find that (68) and (69) gives :

$$(1 + \zeta_n^{\min}) S_n \leq S_{n+1} \leq (1 + \zeta_n^{\max}) S_n. \quad (71)$$

Applying these inequalities recursively and noting that $S_1 = 1$ give us :

$$\prod_{r=1}^{n-1} (1 + \zeta_r^{\min}) \leq S_n \leq \prod_{r=1}^{n-1} (1 + \zeta_r^{\max}). \quad (72)$$

Now, let $\pi = \{A_1, \dots, A_N\}$ be a finite partition of Ω . Then, $\forall A_i \in \pi$, $\exists$ smallest $n_{A_i} < \infty$ and $S_{A_i} \subset \{1, \dots, |\Omega_{n_{A_i}}|\}$ such that $A_i = \bigcup_{k \in S_{A_i}} \text{cyl}(c_{n_{A_i}}^k)$ and we can then write :

$$|A_i\rangle = \left| \bigcup_{k \in S_{A_i}} \text{cyl}(c_{n_{A_i}}^k) \right\rangle = \sum_{k \in S_{A_i}} |c_{n_{A_i}}^k\rangle. \quad (73)$$

And so by the triangle identity we obtain :

$$\| |A_i\rangle \| = \| \sum_{k \in S_{A_i}} |c_{n_{A_i}}^k\rangle \| \leq \sum_{k \in S_{A_i}} \| |c_{n_{A_i}}^k\rangle \|. \quad (74)$$

Let $m = \max(\{n_{A_i}\})$, then :

$$\forall A_i |n_{A_i} < m : |c_{n_{A_i}}^k\rangle = \left| \bigcup_{j(i,k,m)} c_m^{j(k,m)} \right\rangle \Rightarrow \| |c_{n_{A_i}}^k\rangle \| \leq \sum_{j(i,k,m)} \| |c_m^{j(i,k,m)}\rangle \|, \quad (75)$$

(76) and (75) give :

$$\| |A_i\rangle \| \leq \sum_{k \in S_{A_i}} \| |c_{n_{A_i}}^k\rangle \| \leq \sum_{k \in S_{A_i}} \sum_{j(i,k,m)} \| |c_m^{j(i,k,m)}\rangle \|. \quad (76)$$

The set $\{\text{cyl}(c_m^{j(i,k,m)})\}$ is a stage- m cylinder set partition of A_i and the $\{A_i\}$ is a partition of Ω , then :

$$1 = \| |\Omega\rangle \| = \left\| \bigcup_{i=1}^N A_i \right\| = \left\| \sum_{i=1}^N |A_i\rangle \right\| \quad (77)$$

$$\leq \sum_{i=1}^N \| |A_i\rangle \| \leq \sum_{i=1}^N \sum_{k \in S_{A_i}} \sum_{j(i,k,m)} \| |c_m^{j(i,k,m)}\rangle \| = \sum_{i=1}^{|\Omega_m|} \| |c_m^i\rangle \| = S_m \quad (78)$$

And this implies $S_m = \sup_n S_n = |\mu_v|(\Omega)$ and, thanks to (72), this complete the proof. $\square$

7.3 Determination of $\zeta_n^{\min}$ and $\zeta_n^{\max}$ in complex sequential growth [2]

Let first define the function $\lambda(a, b)$ as follows :

$$\lambda(a, b) \equiv \sum_{k=b}^a \binom{a-b}{k-b} t_k. \quad (79)$$

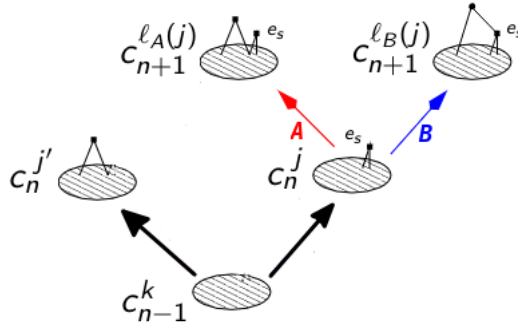


Figure 9: The transitions under consideration

We have defined $\zeta_n^i = \sum_{j(i)} |\alpha(c_n^i \rightarrow c_{n+1}^{j(i)})| - 1$, so :

$$\zeta_n^i = \sum_{(\varpi, m) \in \mathcal{T}(c_n^i)} \frac{|\sum_{l=m}^{\varpi} \binom{\varpi-m}{\varpi-l} t_l|}{|\sum_{j=0}^n \binom{n}{j} t_j|} - 1 = \sum_{(\varpi, m) \in \mathcal{T}(c_n^i)} \frac{|\lambda(\varpi, m)|}{|\lambda(n, 0)|} - 1 \quad (80)$$

where $\mathcal{T}(c_n^i)$ denote the list of (ϖ, m) values for every transition from the parent c_n^i . For any given c_{n-1}^k , we can decompose the set of transitions from c_n^j into two disjoint subset : A is the set of transitions which only involve c_{n-1}^k , so that $e_s = c_n^j \setminus c_{n-1}^k$ is always in the spectator set, B is the set of transitions that always include e_s in the precursor set. Let $\ell_A(j)$ label the type A children of c_n^j , and similarly let $\ell_B(j)$ label the

type B children of c_n^j . For any transition of type A , $c_n^j \rightarrow c_{n+1}^{\ell_A(j)}$, there exists a child $c_n^{j'}$ of c_{n+1}^k such that $c_n^{j(k)} \sim c_{n+1}^{\ell_A(j)} \setminus e_s$. As we can see in the figure 9, the transition $c_n^j \rightarrow c_{n+1}^{\ell_A(j)}$ and $c_{n-1}^k \rightarrow c_n^{j'}$ have the same precursor set. This allows us to express the transition amplitude as :

$$\alpha(c_n^j \rightarrow c_{n+1}^{\ell_A(j)}) = \frac{\lambda(\varpi, m)}{\lambda(n, 0)}, \quad \alpha(c_{n-1}^k \rightarrow c_n^{j'}) = \frac{\lambda(\varpi, m)}{\lambda(n-1, 0)} \quad (81)$$

$$\implies \alpha(c_n^j \rightarrow c_{n+1}^{\ell_A(j)}) = \alpha(c_{n-1}^k \rightarrow c_n^{j'}) \times \frac{\lambda(n-1, 0)}{\lambda(n, 0)}. \quad (82)$$

Summing over all the transitions from c_n^j we find :

$$\begin{aligned} \sum_{\ell} \alpha(c_n^j \rightarrow c_{n+1}^{\ell}(j)) &= \sum_{\ell_A} \alpha(c_n^j \rightarrow c_{n+1}^{\ell_A(j)}) + \sum_{\ell_B} \alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)}) \\ &= \left(\sum_{i(k)} \alpha(c_{n-1}^k \rightarrow c_n^{i(k)}) \right) \frac{\lambda(n-1, 0)}{\lambda(n, 0)} + \sum_B \alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)}). \end{aligned}$$

Which give when we apply the Markov sum rule :

$$1 = \frac{\lambda(n-1, 0)}{\lambda(n, 0)} + \sum_{\ell_B} \alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)}).$$

And since,

$$\begin{aligned} \lambda(n, 0) - \lambda(n-1, 0) &= \sum_{k=0}^n \binom{n}{k} t_k - \sum_{k=0}^{n-1} \binom{n-1}{k} t_k = t_n + \sum_{k=0}^{n-1} \left[\binom{n}{k} - \binom{n-1}{k} \right] t_k \\ &= t_n + \sum_{k=1}^{n-1} \binom{n-1}{k-1} t_k = \sum_{k=1}^n \binom{n-1}{k-1} t_k = \lambda(n, 1), \end{aligned}$$

we see that :

$$\sum_{\ell_B} \alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)}) = \frac{\lambda(n, 1)}{\lambda(n, 0)} \Rightarrow \sum_{\ell_B} |\alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)})| \geq \frac{|\lambda(n, 1)|}{|\lambda(n, 0)|}. \quad (83)$$

By defining $Q_n^j \equiv \zeta_n^j + 1$ we get:

$$Q_n^j = \sum_{(\varpi, m) \in \mathcal{T}(c_n^i)} \frac{|\lambda(\varpi, m)|}{|\lambda(n, 0)|} = Q_{n-1}^{i(j)} \frac{|\lambda(n-1, 0)|}{|\lambda(n, 0)|} + \sum_{\ell_B} |\alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)})| \quad (84)$$

and then

$$Q_n^j \geq Q_{n-1}^{i(j)} \frac{|\lambda(n-1, 0)|}{|\lambda(n, 0)|} + \frac{|\lambda(n, 1)|}{|\lambda(n, 0)|}. \quad (85)$$

Let's focus on the n anti-chain c_n^a , every transition from this causal set satisfy $m = \varpi$ so for the transition $j(a)$ with the number of maximal element m :

$$\alpha(c_n^a \rightarrow c_{n+1}^{j(a)}) = \frac{\lambda(m, m)}{\lambda(n, 0)} = \frac{t_m}{\lambda(n, 0)}. \quad (86)$$

For fixed m there are $\binom{n}{m}$ possible choices of precursor sets for the new element e_{n+1} . So :

$$Q_n^a = \frac{\sum_{k=0}^n \binom{n}{k} |t_k|}{|\lambda(n, 0)|} \implies (55). \quad (87)$$

For every causal set c_n^j the Markov sum rule and the definition (79) yields

$$\sum_{(\varpi, m) \in \mathcal{T}(c_n^j)} \lambda(\varpi, m) = \lambda(n, 0) = \sum_{k=0}^n \binom{n}{k} t_k. \quad (88)$$

Because, $|\mathcal{T}(c_n^j)| < n = |\mathcal{T}(c_n^a)|$ we have :

$$\sum_{(\varpi, m) \in \mathcal{T}(c_n^j)} |\lambda(\varpi, m)| \leq \sum_{k=0}^n \binom{n}{k} |t_k|. \quad (89)$$

Thanks to (84) and (87) we can write $Q_n^j \leq Q_n^a$ so $\zeta_n^j \leq \underline{\zeta_n^a} = \zeta_n^{\max}$.

Let's now prove by induction that $\forall n, \forall j \in \{1, \dots, |\Omega_n|\}$ $Q_n^j \geq Q_n^c$:

For $n = 0, 1$, the precursor set is either the empty set or a single element so $Q_n^j = Q_n^c$. Now let's assume that $Q_{n-1}^j \geq Q_{n-1}^c$ knowing this and (85) it yields :

$$Q_n^j \geq Q_{n-1}^c \frac{|\lambda(n-1, 0)|}{|\lambda(n, 0)|} + \frac{|\lambda(n, 1)|}{|\lambda(n, 0)|}. \quad (90)$$

From c_n^c , the only possible transition of type (B) is that with e_n as the (unique) maximal element of the precursor set : $c_n^c \rightarrow c_{n+1}^c$ so with (84) :

$$Q_n^c = Q_{n-1}^c \frac{|\lambda(n-1, 0)|}{|\lambda(n, 0)|} + \sum_{\ell_B} |\alpha(c_n^j \rightarrow c_{n+1}^{\ell_B(j)})| = Q_n^c = Q_{n-1}^c \frac{|\lambda(n-1, 0)|}{|\lambda(n, 0)|} + \frac{|\lambda(n, 1)|}{|\lambda(n, 0)|}. \quad (91)$$

This proves that $\underline{\zeta_n^c} = \zeta_n^{\min}$ and by using (91) iteratively, we show (56). $\square$

7.4 In $\mathbb{CP}$, $p \notin \mathbb{R} \implies \mu_v$ is not of bounded variation .

Let's take $p = 1 - q \notin \mathbb{R}$, we have, then :

$$p + q = 1 \text{ and } |p| + |q| = 1 + \zeta \text{ with } \zeta > 0. \quad (92)$$

By definition of the variation of a measure :

$$|\mu_v|(A) \equiv \sup_{\pi} \sum_{a_i \in \pi} \|\mu_v(a_i)\| \geq \sum_{a_i \in \pi} \|\mu_v(a_i)\| \quad (93)$$

with π denoting the partitions of A. Since we have :

$$\|\mu_v(a_i)\| = \| |a_i\rangle \| = \sqrt{\langle a_i | a_i \rangle} = \sqrt{D(a_i, a_i)} = \sqrt{|\alpha(a_i)|^2} = |\alpha(a_i)| = \|\alpha(a_i)\|, \quad (94)$$

we can write $\forall A \in \mathfrak{A}$:

$$|\mu_v|(A) \equiv \sup_{\pi} \sum_{a_i \in \pi} \|\mu_v(a_i)\| = \sup_{\pi} \sum_{a_i \in \pi} |\alpha(a_i)| = |\alpha|(A). \quad (95)$$

In the complex case, the α are amplitude of probability. We can recover the classical probability by taking the square modulus. Then, thanks to (20) we can write :

$$\alpha(c_n^c) = p^{n-1} \implies |\alpha|(\Omega) > |p|^{n-1}, \quad (96)$$

$$\alpha(c_n^a) = q^{n(n-1)} \implies |\alpha|(\Omega) > |q|^{n(n-1)}. \quad (97)$$

If we have $|p| > 1$ or $|q| > 1$, (96) and (97) implies that μ_v is not of bounded variation. We need to show now that it is still it in the case where $|p| < 1$ and $|q| < 1$. We will suppose it in the following.

We will denote by $c_{n+1}^{j_1(i)}$ the child of c_n^i with the label j_1 (J_1^i is the number of such children), $c_{n+2}^{j_2 j_1(i)}$ the child of $c_{n+1}^{j_1(i)}$ with the label j_2 ($J_2^i(j_1)$ is the number of such children) and so on. The set $\Pi \equiv \{c_{n+s}^{j_s \dots j_2 j_1(i)}\}$ of $n+s$ element descendants of c_n^i provides a disjoint partition of $\text{cyl}(c_n^i)$. So :

$$|\alpha|(c_n^i) \geq \sum_{j_1=1}^{J_1^i} \sum_{j_2=1}^{J_2^i(j_1)} \cdots \sum_{j_s=1}^{J_s^i(j_{s-1} j_{s-2} \dots j_1)} |\alpha|(c_{n+s}^{j_s \dots j_2 j_1(i)}), \quad (98)$$

where

$$\alpha(c_{n+s}^{j_s \dots j_2 j_1(i)}) = \alpha(c_n^i) \alpha(c_n^i \rightarrow c_{n+1}^{j_1(i)}) \alpha(c_{n+1}^{j_1(i)} \rightarrow c_{n+2}^{j_2 j_1(i)}) \dots \alpha(c_{n+s-1}^{j_{s-1} \dots j_2 j_1(i)} \rightarrow c_{n+s}^{j_s j_{s-1} \dots j_2 j_1(i)}). \quad (99)$$

Thus,

$$|\alpha|(c_n^i) \geq |\alpha|(c_n^i) \sum_{j_1=1}^{J_1^i} |\alpha|(c_n^i \rightarrow c_{n+1}^{j_1(i)}) \sum_{j_2=1}^{J_2^i(j_1)} |\alpha|(c_{n+1}^{j_1(i)} \rightarrow c_{n+2}^{j_2 j_1(i)}) \cdots \sum_{j_s=1}^{J_s^i(j_{s-1} j_{s-2} \dots j_1)} |\alpha|(c_{n+s-1}^{j_{s-1} \dots j_2 j_1(i)} \rightarrow c_{n+s}^{j_s j_{s-1} \dots j_2 j_1(i)}). \quad (100)$$

Now we will isolate one sum :

$$\sum_{j_s} |\alpha|(c_{n+s-1}^{j_{s-1} \dots j_2 j_1(i)} \rightarrow c_{n+s}^{j_s j_{s-1} \dots j_2 j_1(i)}) = \sum_{j_s} |p|^{u_{j_s}} |q|^{v_{j_s}} = (1 + \zeta - |q|)^m + \sum_{i \in J'_s} (1 + \zeta - |q|)^{u_i} |q|^{v_i} \quad (101)$$

where J'_s is the index set which exclude $(u, v) = (m, 0)$ where u is the number of links and v the number of spacelike relations between the new element and the elements of the parent causal set. Using the Binomial theorem and denoting J''_s the index set which exclude $(u, v) = (m, 0)$ and $(u, v) = (0, n+s-1)$ it yields :

$$\begin{aligned} \sum_{j_s} |\alpha|(c_{n+s-1}^{j_{s-1} \dots j_2 j_1(i)} \rightarrow c_{n+s}^{j_s j_{s-1} \dots j_2 j_1(i)}) &= (1 - |q|)^m + \sum_{i \in J'_s} (1 - |q|)^{u_i} |q|^{v_i} \\ &+ \binom{m}{1} (1 - |q|)^{m-1} \zeta + \binom{m}{2} (1 - |q|)^{m-2} \zeta^2 + \cdots + \zeta^m \\ &+ \sum_{k \in J''_s} \left(\binom{u_k}{1} (1 - |q|)^{u_k-1} \zeta + \binom{u_k}{2} (1 - |q|)^{u_k-2} \zeta^2 + \cdots + \zeta^{u_k} \right) |q|^{v_k}. \end{aligned}$$

By focusing on the terms linear in ζ , we can see that those are greater than 1 for $|q| < 1$. We have :

$$\frac{(1 - |q|)^m + \sum_{i \in J''_s} (1 - |q|)^{u_i} |q|^{v_i}}{1 - |q|} + \left((m-1)(1 - |q|)^{m-1} + \sum_{j \in J''_s} \left((u_j - 1)(1 - |q|)^{u_j-1} |q|^{v_j} \right) \right) > \frac{1 - |q|^{n+s-1}}{1 - |q|} \geq 1 \quad (102)$$

Because $\sum_{i \in J_s} (1 - |q|)^{u_i} |q|^{v_i} = 1$ and all the terms in the sum are positive. We find then $\forall s$,

$$\sum_{j_s} |\alpha|(c_{n+s-1}^{j_{s-1} \dots j_2 j_1(i)} \rightarrow c_{n+s}^{j_s j_{s-1} \dots j_2 j_1(i)}) \geq 1 + \zeta, \quad (103)$$

from (100), this implies :

$$|\alpha|(c_n^i) \geq |\alpha|(c_n^i) \times (1 + \zeta)^s. \quad (104)$$

Since s can be made arbitrarily large, this means that $|\alpha|(c_n^i)$ is not bounded. $\square$

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